

EPHRATA, WA, USA

WMO: 727900

Lat: 47.308N

Lon: 119.515W

Elev: 1252

StdP: 14.04

Time Zone: -8.00 (NAP)

Period: 01-19

WBAN: 24141

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	9.0	13.6	-1.3	5.4	13.0	3.6	6.9	16.4	25.0	33.7	20.7	34.2	6.6	340	0.574

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.4	97.9	64.6	94.7	63.8	91.4	62.7	66.8	92.8	65.2	90.3	63.8	87.8	9.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.9	74.9	71.0	55.7	69.1	70.9	53.7	64.2	71.2	31.9	93.2	30.6	89.9	29.5	87.3	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	20.1	17.9	0.8	103.0	6.4	2.9	-3.8	105.1	-7.5	106.8	-11.1	108.4	-15.8	110.6
			0.1	69.0	6.2	2.3	-4.4	70.7	-8.0	72.0	-11.5	73.3	-16.0	75.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.9	29.9	34.8	43.2	50.9	60.8	68.0	76.4	74.3	64.7	51.2	38.1	29.7	(6)
	DBStd	17.70	8.31	7.17	6.63	6.33	7.35	7.28	6.46	5.98	6.97	7.05	8.04	7.41	(7)
	HDD50	2399	622	425	223	63	5	0	0	0	2	66	363	630	(8)
	HDD65	5681	1087	844	677	425	174	48	2	4	88	429	808	1095	(9)
	CDD50	3107	0	0	11	89	340	542	819	754	443	104	5	0	(10)
	CDD65	913	0	0	0	1	43	139	356	293	79	2	0	0	(11)
	CDH74	12049	0	0	0	45	610	1759	4812	3753	1034	36	0	0	(12)
	CDH80	5696	0	0	0	6	197	751	2527	1828	381	6	0	0	(13)
(14) Wind	WSAvg	7.7	6.3	7.4	8.0	8.9	8.7	9.0	9.0	8.0	7.6	7.0	6.6	6.3	(14)
(15) Precipitation	PrecAvg	8.0	0.9	0.6	0.6	0.6	0.8	0.7	0.3	0.3	0.2	0.7	0.9	1.2	(15)
	PrecMax	12.4	2.6	1.5	1.7	1.2	1.9	1.7	1.9	1.2	0.8	2.6	1.9	3.3	(16)
	PrecMin	4.4	0.2	0.0	0.0	0.1	0.1	0.1	0.0	0.0	0.0	0.1	0.0	0.0	(17)
	PrecStd	2.0	0.7	0.4	0.4	0.3	0.6	0.5	0.4	0.3	0.2	0.5	0.6	0.9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.4	57.0	66.7	79.5	90.9	99.2	102.2	100.2	94.2	79.1	61.9	50.7	(19)
		MCWB	42.8	46.7	51.7	56.0	60.5	66.1	66.4	64.8	63.0	59.3	52.0	44.1	(20)
	2%	DB	45.6	52.3	62.7	73.1	85.6	93.0	98.6	96.9	88.7	72.2	56.8	45.8	(21)
		MCWB	40.2	43.6	49.1	54.0	60.1	63.8	65.1	64.4	61.3	55.6	48.1	40.8	(22)
	5%	DB	42.4	48.8	58.7	68.4	80.9	88.1	95.8	93.2	84.3	67.8	52.8	42.0	(23)
		MCWB	38.3	41.3	46.7	51.5	57.6	61.8	64.1	63.3	59.7	52.7	45.5	37.9	(24)
	10%	DB	39.3	45.4	55.3	64.1	76.4	83.7	92.6	89.8	80.0	63.8	49.5	39.0	(25)
		MCWB	36.5	39.2	44.9	48.8	55.6	59.6	63.1	62.1	57.8	50.8	43.4	36.0	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.0	48.4	53.5	58.2	63.2	67.9	69.1	69.7	65.1	60.7	52.9	45.8	(27)
		MCDB	48.8	55.4	63.0	76.5	85.9	94.7	96.0	93.6	88.6	73.5	59.0	49.0	(28)
	2%	WB	40.9	44.7	50.2	55.0	60.7	65.1	67.3	66.1	62.7	57.1	49.8	41.8	(29)
		MCDB	44.9	50.5	60.1	70.6	80.4	90.2	94.0	90.7	84.8	68.9	55.4	44.9	(30)
	5%	WB	38.7	42.2	48.0	52.5	59.0	63.0	65.7	64.5	60.8	54.5	47.0	38.6	(31)
		MCDB	41.6	47.2	57.1	66.8	76.9	85.9	92.1	88.1	81.0	64.8	51.4	41.3	(32)
	10%	WB	36.9	40.3	46.0	50.3	57.1	60.7	64.0	63.2	59.0	52.2	44.4	36.4	(33)
		MCDB	39.1	44.9	53.8	62.1	73.0	81.1	89.2	86.2	77.3	61.7	48.5	38.7	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	10.5	15.4	20.6	22.7	23.7	23.6	26.4	26.5	25.5	21.3	15.1	10.7	(35)
		MCDBR	14.8	20.2	24.8	28.3	29.1	28.7	29.5	30.1	30.9	26.1	19.0	15.2	(36)
	5% WB	MCWBR	10.4	13.0	14.2	13.8	11.4	11.1	10.9	11.4	13.0	13.3	12.2	10.8	(37)
		MCWBR	13.3	18.4	23.4	27.3	26.2	27.4	27.8	27.9	28.3	23.2	17.5	13.2	(38)
(40) Clear-Sky Solar Irradiance	taub		0.277	0.283	0.294	0.318	0.333	0.335	0.336	0.334	0.322	0.303	0.288	0.273	(40)
		taud	2.506	2.559	2.572	2.473	2.446	2.467	2.463	2.460	2.496	2.544	2.545	2.515	(41)
	Ebn at Noon		263	286	298	298	295	294	292	289	283	273	256	251	(42)
		Edh at Noon	20	23	26	32	34	34	34	32	28	23	19	17	(43)
(44) All-Sky Solar Radiation	RadAvg		344	700	1117	1654	2017	2213	2402	2010	1480	909	456	285	(44)
	RadStd		40	89	77	101	124	124	94	110	82	75	51	37	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	-0.79	N/A	N/A	N/A	+171	+85

Nomenclature: See separate page